An Open API specification details what the API structure looks like. It defines the available endpoints and the properties they can provide. An Open API specification can be used by teams to author integration tests. However, relying on an Open API specification doesn't enforce the prevention of integration bugs automatically because teams can deploy multiple changes to test environments. The contract generated through contract testing can prevent integration bugs because it informs consumers and providers when changes are detected, especially when integrated into a CI pipeline, even before deploying the changes to a test environment.

Integration tests are also often authored by just one team. This could be the team maintaining an API or the team using the API. This can cause challenges when it comes to sharing resources and knowledge across different services or applications. Meanwhile, in consumer-driven contract testing, the tests are authored by the team using the actual API service (messages are also supported, which we'll discuss in chapter 8), and the same set of tests are also verified by the team maintaining the API service.

1.4.1 Contract testing from the consumer side

When working with consumer-driven contract testing, most of the test writing will happen on the consumer side because they will define the contract. Let's consider an example. Imagine a frontend engineer working on the web application is working on a new user page. The frontend engineer writes a contract test using a contract testing tool called Pact (https://pact.io/) to ensure that the user page doesn't have any unexpected problems related to the data it's supposed to display. The process of writing the contract test from the consumer side can be visualized in figure 1.5.